

chapter is applied) and therefore miss the impact of sequence-of-returns risk. The basic approach of assuming a fixed return reflecting the best guess one can make about future market returns leads to a retirement plan with only a 50 percent chance to work. The outcomes are too optimistic and could lead a retiree down an unsustainable path.

A second alternative is to use “historical simulations” from rolling periods of the data. The outcomes for a thirty-year retirement could be studied using each possible thirty-year period available in the data. It was with this approach that William Bengen developed the 4 percent rule.

Third, Monte Carlo simulations provide an alternative that is now widely used in financial planning software. The simulations are used to develop sequences of random market returns fitting predetermined characteristics, in order to test how financial plans will perform in a wider variety of good and bad market environments. The use of Monte Carlo tools has increased considerably over the past decade, which can likely be attributed to lower computing costs, increased recognition that returns are random, and desires to provide more robust financial plans. A thousand or more simulations could be created to test the robustness of a retirement plan in many market environments.

Monte Carlo simulations can be created for different asset classes or for an overall portfolio. With the asset class approach, one defines the arithmetic average return, the volatility for that return, and the correlations with other asset classes. Random draws are then taken from multivariate normal or lognormal distributions sharing these characteristics. By combining the arithmetic mean with volatility, the resulting simulated returns will display the appropriate compounded return. Historical data is commonly used to set these input characteristics. Most financial planning software works in this way.

With Monte Carlo simulations, though, the sky is the limit about what types of assumptions can be used to prepare the assumptions guiding the simulations. For instance, one could assume “fat tail” distributions that allow for more extreme market outcomes, or one could incorporate serial